

Applied Mathematics for Chemists

(화학수학)

Lecture 3

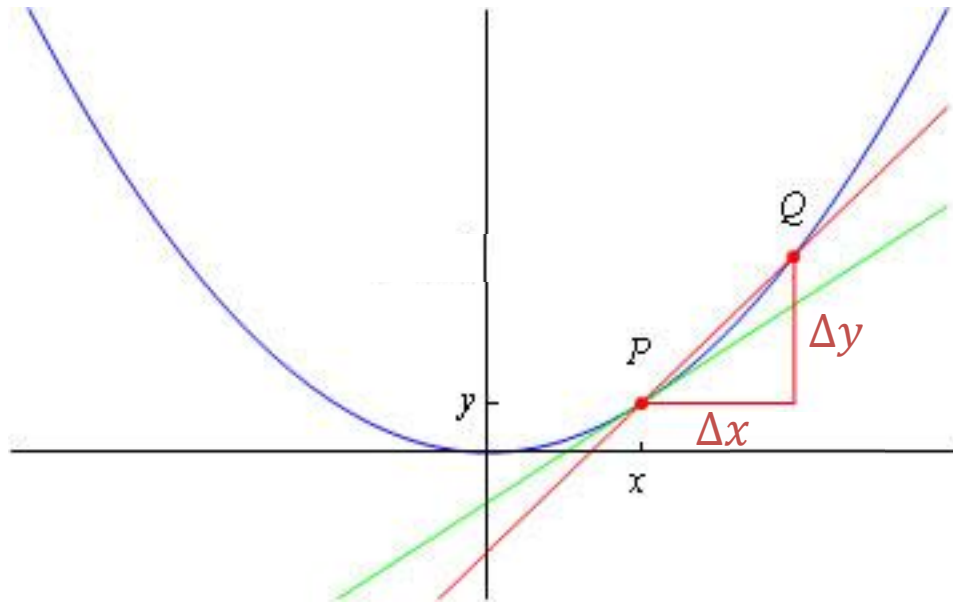
Multivariate Differential Calculus

Today's Class

- [Review] Integration
- Multivariate functions
- Partial derivatives
 - Fundamental equation of differential calculus
 - Change of variables
- Useful partial derivative identities

Differentiation

“The slope of a function”

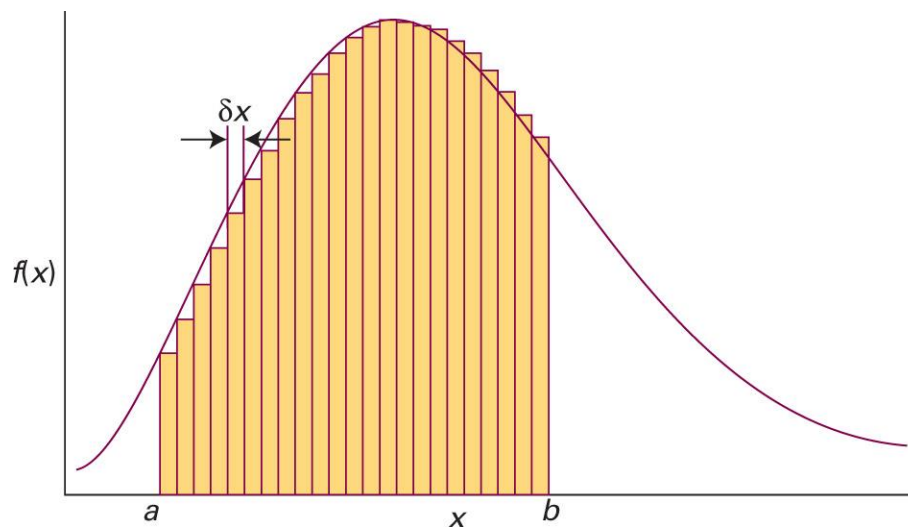


$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$

(Image: http://wwwf.imperial.ac.uk/metric/metric_public/differentiation/index.html)

Integration

“The areas under curves”



$$\lim_{\delta x \rightarrow 0} \sum_i f(x_i) \delta x = \int_a^b f(x) dx$$

Indefinite Integrals (부정적분)

Consider an integral

$$\int_a^{x'} f(x) dx = F(x') + C$$

constant of
integration
(적분상수)

where x' is a variable and a is an arbitrary constant.

Generally, it is written as

$$\int f(x) dx = F(x) + C$$

Calculation of Indefinite Integrals



When evaluating the integral $F(x) = \int f(x)dx$,
find a function $F(x)$ whose derivative is $f(x)$:

$$f(x) = \frac{dF(x)}{dx}$$

Some Examples of Derivatives

Integration



Function, $y = y(x)$	Derivative, $dy/dx = y'(x)$
ax^n	anx^{n-1}
ae^{bx}	abe^{bx}
a	0
$a \sin(bx)$	$ab \cos(bx)$
$a \cos(bx)$	$-ab \sin(bx)$
$a \ln(x)$	a/x
$a \tan(x)$	

Check Appendix E!

Calculation of Definite Integrals

If the indefinite integral of $f(x)$ is $F(x)$,

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

No constant of integration!

Techniques of Integration

Method of substitution (변수치환법)

Integration by parts (부분적분법)

Method of partial fractions (부분분수법)

Method of Substitution

1. Define a new variable
2. Calculate its infinitesimal change and the limits
3. Rewrite the integral in terms of the new variable

Example 7.9

Evaluate the integral

$$\int_0^{\infty} x e^{-x^2} dx.$$

Let $u = x^2$. Then,

$$du = 2x dx$$

$$x = 0 \rightarrow u = 0$$

$$x = \infty \rightarrow u = \infty$$

Hence, substitution leads to

$$\int_0^{\infty} \frac{1}{2} e^{-u} du = \left[-\frac{1}{2} e^{-u} \right]_0^{\infty} = \frac{1}{2}$$

Integration by Parts

If u and v are both functions of x ,

$$\frac{d(uv)}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

Integrating both sides,

$$uv = \int v \frac{du}{dx} dx + \int u \frac{dv}{dx} dx + C$$

which gives

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx + C$$

Integration by Parts

For indefinite integrals,

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx + C$$

For definite integrals,

$$\int_a^b u \frac{dv}{dx} dx = uv \Big|_a^b - \int_a^b v \frac{du}{dx} dx$$

Example 7.11

Find the definite integral

$$\int_0^{\pi} x \sin x \, dx.$$

Let $u = x$ and $dv/dx = \sin x$. Then, $v = -\cos x$.

$$\int_0^{\pi} x \sin x \, dx = \int_0^{\pi} u \frac{dv}{dx} dx = uv \Big|_0^{\pi} - \int_0^{\pi} v \frac{du}{dx} dx$$

Hence, substitution leads to

$$x(-\cos x) \Big|_0^{\pi} - \int_0^{\pi} (-\cos x) \, dx = \pi + (\sin x) \Big|_0^{\pi} = \pi$$

Method of Partial Fractions

If $P(x)$ and $Q(x)$ are polynomials, and $Q(x)$ can be factorized as

$$Q(x) = (a_1x + b_1)(a_2x + b_2) \cdots (a_nx + b_n),$$

$P(x)/Q(x)$ can be decomposed as

$$\frac{P(x)}{Q(x)} = \frac{A_1}{a_1x + b_1} + \frac{A_2}{a_2x + b_2} + \cdots + \frac{A_n}{a_nx + b_n}$$

Simpler combination of fractions

Example 7.12

Decompose the integral

$$\int \frac{6x - 30}{x^2 + 3x + 2} dx.$$

The denominator is decomposed as $x^2 + 3x + 2 = (x + 2)(x + 1)$. Thus,

$$\frac{6x - 30}{x^2 + 3x + 2} = \frac{A_1}{x + 2} + \frac{A_2}{x + 1}$$

Multiplying both sides by $(x + 2)(x + 1)$,

$$6x - 30 = A_1(x + 1) + A_2(x + 2)$$

$$6x - 30 = (A_1 + A_2)x + (A_1 + 2A_2)$$

Since $A_1 + A_2 = 6$ and $A_1 + 2A_2 = -30$, $A_1 = 42$ and $A_2 = -36$. Hence,

$$\int \frac{6x - 30}{x^2 + 3x + 2} dx = \int \frac{42}{x + 2} dx - \int \frac{36}{x + 1} dx.$$

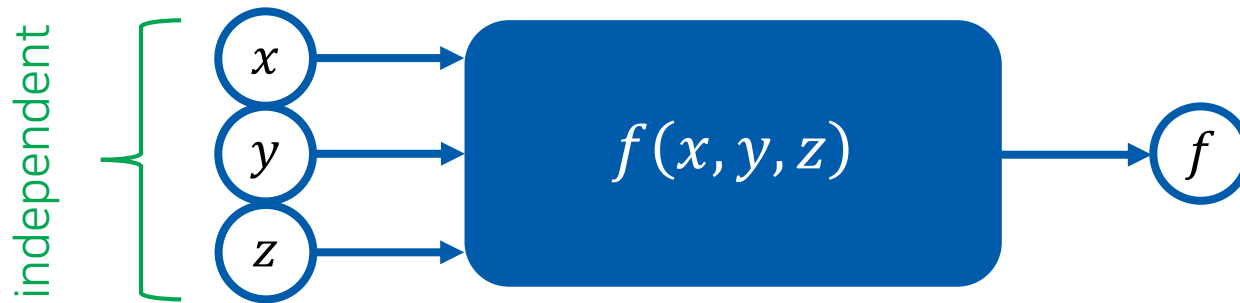


8. Differential Calculus with Several Independent Variables

(8장 다변수 미분)

Function of Several Independent Variables

(다변수함수)



Example: $f(x, y, z) = x^2 + xy + 3z$

$$f(1, 1, 1) = 1 + 1 + 3 = 5$$

$$f(-1, 1, 0) = 1 - 1 + 0 = 0$$

⋮

Examples from Physical Chemistry

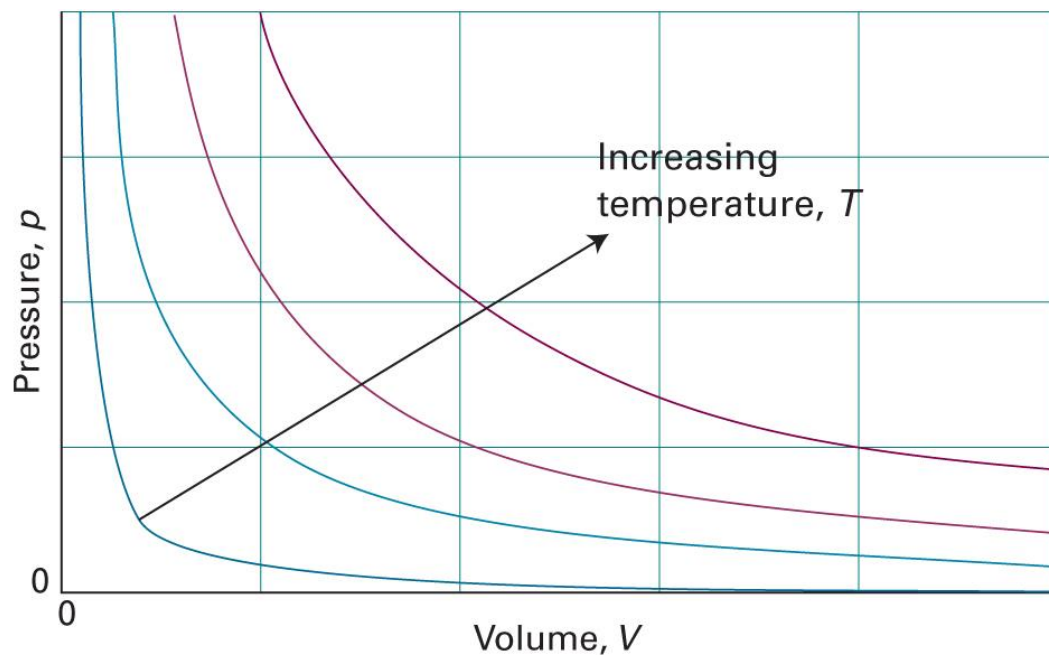
- Thermodynamic properties
 - Pressure $P = P(T, V, n)$
 - Internal energy $U = U(T, V, n)$
- Wave function $\psi = \psi(x, y, z)$
- Reaction rate $r = r([A], [B])$
- piecewise continuous and single-valued

Example: Ideal Gas Equation

$$P = P(T, V, n) = \frac{nRT}{V}$$

Visualization: Single Independent Variable

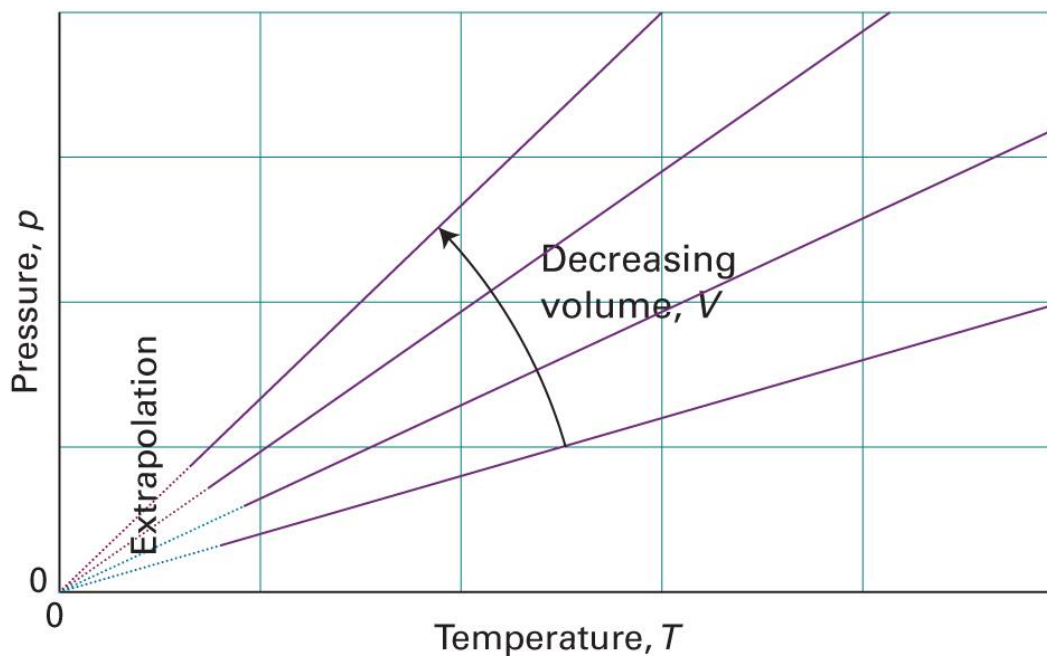
$$P = P(T, V, n) = \frac{nRT}{V}$$



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1A.2)

Visualization: Single Independent Variable

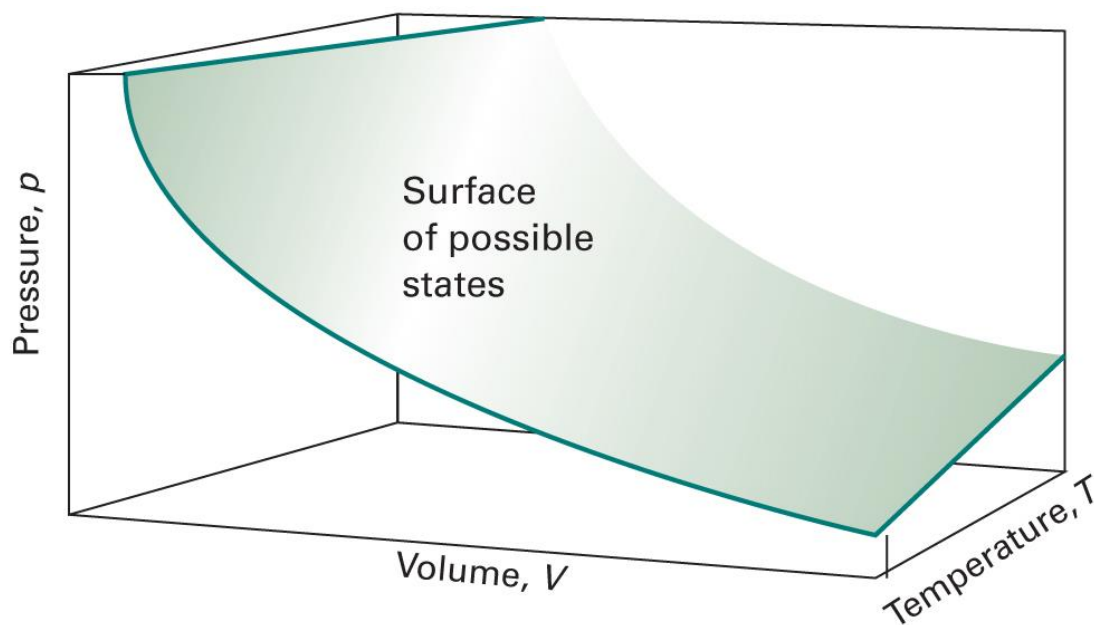
$$P = P(T, V, n) = \frac{nRT}{V}$$



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1A.5)

Visualization: Two Independent Variables

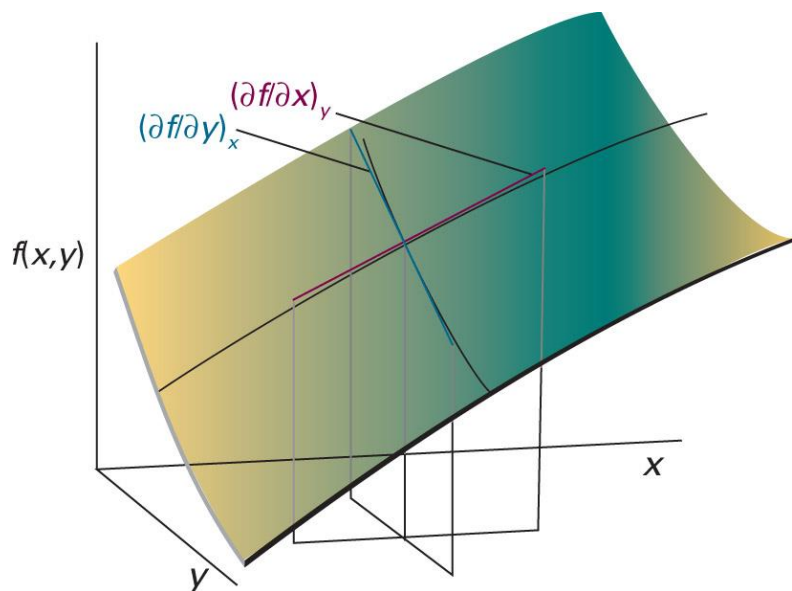
$$P = P(T, V, n) = \frac{nRT}{V}$$



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1A.6)

Partial Derivatives (편미분)

“the slope of the function with respect to one of the variables,
all the other variables being held constant”



$$\left(\frac{\partial f}{\partial x}\right)_y = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\left(\frac{\partial f}{\partial y}\right)_x = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Calculation of Partial Derivatives

Treat fixed variables like **ordinary constants**

$$\left(\frac{\partial P}{\partial V}\right)_{n,T} = \left(\frac{\partial}{\partial V} \left[\frac{nRT}{V} \right]\right)_{n,T} = -\frac{nRT}{V^2}$$

$$\left(\frac{\partial P}{\partial T}\right)_{n,V} = \left(\frac{\partial}{\partial T} \left[\frac{nRT}{V} \right]\right)_{n,V} = \frac{nR}{V}$$

$$\left(\frac{\partial P}{\partial n}\right)_{T,V} = \left(\frac{\partial}{\partial n} \left[\frac{nRT}{V} \right]\right)_{T,V} = \frac{RT}{V}$$

Infinitesimal Changes

For $P = P(T, V, n)$,

$$dP = \left(\frac{\partial P}{\partial V} \right)_{n,T} dV \quad (n \text{ and } T \text{ fixed})$$

$$dP = \left(\frac{\partial P}{\partial V} \right)_{n,T} dV + \left(\frac{\partial P}{\partial T} \right)_{n,V} dT \quad (n \text{ fixed})$$

$$dP = \left(\frac{\partial P}{\partial V} \right)_{n,T} dV + \left(\frac{\partial P}{\partial T} \right)_{n,V} dT + \left(\frac{\partial P}{\partial n} \right)_{T,V} dn$$



(total) differential of P (P 에 대한 전미분)

Fundamental Equation of Differential Calculus

(미분학의 기본 방정식)

- n independent variables: $x_1, x_2, \dots, x_n$
- dependent variable: y

The differential of y is

$$dy = \sum_{i=1}^n \left(\frac{\partial y}{\partial x_i} \right)_{x'} dx_i$$

where x' means keeping all of the variables except for x_i fixed

Case of Ideal Gas

$$dP = \left(\frac{\partial P}{\partial V} \right)_{n,T} dV + \left(\frac{\partial P}{\partial T} \right)_{n,V} dT + \left(\frac{\partial P}{\partial n} \right)_{T,V} dn$$

$$dP = -\frac{nRT}{V^2} dV + \frac{nR}{V} dT + \frac{RT}{V} dn$$

Approximately,

$$\Delta P \approx -\frac{nRT}{V^2} \Delta V + \frac{nR}{V} \Delta T + \frac{RT}{V} \Delta n$$

Example 8.1

Calculate the approximate change in pressure of an ideal gas if

- the volume is changed from 20.000 L to 19.800 L,
- the temperature is changed from 298.15 K to 299.00 K, and
- the amount of gas in moles is changed from 1.0000 mol to 1.0015 mol.

Compare the result with the correct value of the pressure change.

$$\Delta P \approx -\frac{nRT}{V^2} \Delta V + \frac{nR}{V} \Delta T + \frac{RT}{V} \Delta n$$

In SI units, $R = 8.3145 \text{ J K}^{-1} \text{ mol}^{-1}$, and

$V = 0.020000 \text{ m}^3$, $T = 298.15 \text{ K}$, $n = 1.0000 \text{ mol}$ (initial values);

$\Delta V = -2.00 \times 10^{-4} \text{ m}^3$, $\Delta T = 0.85 \text{ K}$, $\Delta n = 1.5 \times 10^{-3} \text{ mol}$

Substitution leads to $\Delta P = 1.779 \times 10^3 \text{ N m}^{-2} = 1.779 \times 10^3 \text{ Pa}$



Example 8.1

Calculate the approximate change in pressure of an ideal gas if

- the volume is changed from 20.000 L to 19.800 L,
- the temperature is changed from 298.15 K to 299.00 K, and
- the amount of gas in moles is changed from 1.0000 mol to 1.0015 mol.

Compare the result with the correct value of the pressure change.

Approximate value: $\Delta P = 1.779 \times 10^3 \text{ N m}^{-2} = 1.779 \times 10^3 \text{ Pa}$

Correct value:

$$\Delta P = P_2 - P_1 = \frac{n_2 R T_2}{V_2} - \frac{n_1 R T_1}{V_1}$$

Substitution leads to $\Delta P = 1.797 \times 10^3 \text{ Pa}$. The error is about 1%.

In General

For $U = U(P, T, n)$,

$$dU = \left(\frac{\partial U}{\partial P} \right)_{T,n} dP + \left(\frac{\partial U}{\partial T} \right)_{P,n} dT + \left(\frac{\partial U}{\partial n} \right)_{P,T} dn$$

Approximately,

$$\Delta U \approx \left(\frac{\partial U}{\partial P} \right)_{T,n} \Delta P + \left(\frac{\partial U}{\partial T} \right)_{P,n} \Delta T + \left(\frac{\partial U}{\partial n} \right)_{P,T} \Delta n$$

Example 8.2

For a sample of 1.000 mol of liquid CCl_4 at a temperature of 20°C and a pressure of 1.000 atm, the following experimental values are available:

$$(\partial U/\partial T)_{P,n} = 129.4 \text{ J K}^{-1}, (\partial U/\partial P)_{T,n} = 8.51 \times 10^{-4} \text{ J atm}^{-1}$$

Estimate the change in the energy of 1.000 mol of CCl_4 if its temperature is changed from 20.0 to 40.0°C and its pressure from 1.0 to 100.0 atm.

Since n is fixed,

$$\Delta U \approx \left(\frac{\partial U}{\partial P}\right)_{T,n} \Delta P + \left(\frac{\partial U}{\partial T}\right)_{P,n} \Delta T + \left(\frac{\partial U}{\partial n}\right)_{P,T} \Delta n = \left(\frac{\partial U}{\partial P}\right)_{T,n} \Delta P + \left(\frac{\partial U}{\partial T}\right)_{P,n} \Delta T$$

From the given values, $\Delta T = 20.0 \text{ K}$ and $\Delta P = 99.0 \text{ atm}$

Substitution leads to $\Delta U = 2588 \text{ J} + 0.084 \text{ J} = 2588 \text{ J}$